MIL-F-80079B
22 February 1979
SUPERSEDING
MIL-F-80079A
1 August 1973

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, ELECTRIC CONTROLLED ATMOSPHERE, HORIZONTAL BOX DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heat-treating furnaces of the electrically heated, controlled atmosphere, horizontal box design.

1.2 Classification. This specification covers furnaces of the following classes. The class required shall be as specified (see 6.2.1).

1.2.1 Classes.

Class 1 - 1400°F to 2000°F operating temperature range
Class 2 - 1600°F to 2500°F operating temperature range
Class 3 - 2000°F to 2750°F operating temperature range

* 1.2.2 Sizes. The size of the furnace chamber shall be specified (see 6.2.1). The minimum size of the furnace chamber shall be not less than 6 inches in width x 12 inches in length x 6 inches in height. Size changes shall increase in increments of 3 inches in width or height and 6 inches in length up to a chamber size of 18 W x 30 L x 18 H. Chambers larger than 18 W x 30 L x 18 H shall increase in increments of 6 inches in any dimension as required. The inside dimensions of the heating chamber shall be not less than the dimensions specified. The door opening size shall be not less than width and height dimensions specified for the size furnace.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments	Occupational Safety and Health Administration Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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JOINT INDUSTRIAL COUNCIL (JIC)

EGP-1 Electrical Standards for General Purpose Machine Tools

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook for Industrial Loss Prevention

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston - Providence Turnpike, Norwood, MA 02062.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C96.1 Temperature Measurement Thermocouples
Z210.1 Metric Practice (ASTM-E-380)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association standards and specifications are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete furnace for first article inspection and approval (see 4.2 and 6.3).

3.2 Design. Furnaces and equipment provided in accordance with this specification shall be new and of the manufacturer's current design. The furnaces shall be of the horizontal box design, having a hearth to support the charge and a single opening located in the end of the chamber for inserting and removing the charge in a horizontal plane. Unless otherwise specified, the furnaces shall employ the radiation method of heating with the charge directly exposed to the heat source. When required (see 6.2.1), the furnaces shall be provided with a retort to separate the heating elements from the atmosphere. Each furnace shall be of gas-tight construction, and designed and equipped for operation with the type of controlled atmosphere specified (see 6.2.1). The furnaces shall be floor mounted and shall not require either excavation or a foundation.

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* 3.2.1 Safety and health requirements. Each furnace shall be provided with safety features and devices for the protection of personnel, equipment and the work being processed. In addition to other safety requirements specified herein, the safety system of the furnace shall conform to the applicable requirements of FM "Handbook of Industrial Loss Prevention". The furnace shall comply with the standards promulgated under Occupational Safety and Health Act (OSHA) applicable to the furnace itself. Additional requirements for safety and health shall be as specified (see 6.2.1 and 6.4).

3.2.2 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack-seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material or equipment. Lubrication points shall be readily accessible, and identified in the operators manual with type of lubricant and instructions for application.

3.2.3 Material. All material used in fabrication of the furnaces and related equipment shall be of the quality necessary to produce furnaces to meet the requirements described herein. Materials that are exposed to the internal environment of the furnace chamber shall be compatible with the type of atmosphere and temperature conditions specified for the furance.

3.2.4 Reclaimed materials. The furnace may contain reclaimed materials provided such materials will not jeopardize the furnace intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the furnaces.

3.2.5 Interchangeability. All parts subject to replacement shall be remanufactured to definite dimensions and tolerance standards, which will permit installation of replacement parts without modification to parts or equipment.

3.2.6 Accessibility. All parts subject to wear, breakage, or distortion and all parts which require periodic lubrication or maintenance shall be accessible for adjustment, replacement and lubrication, as applicable.

3.3 Construction. The furnace shall be constructed so that when installed and connected to the specified power source, it can be used for any purpose described herein. All parts and accessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

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3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. The furnace shall be primed and painted with heat resisting paint in accordance with the manufacturer's standard commercial practice.

* 3.3.6 Threads. All threaded parts shall conform to FED-STD-H28.

3.3.7 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

* 3.3.8 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the furnace. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, temperatures and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI Z210.1.

* 3.4 Electrical equipment. The furnaces shall be furnished with electrical motors, contactors, controls, relays, transformers, signal lights, and switches, pre-wired where possible for operation upon connection to the source of power. A fusible disconnect switch or circuit breaker shall be provided to disconnect all power to the furnace. All electrical equipment shall conform to the JIC Electrical Standard for General Purpose Machine Tools.

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3.4.1 Power supply. The furnaces shall operate from a single source primary power supply. A stepdown transformer shall be provided for the 120 volt control system power supply. The furnace shall be designed for operating from the power source specified (see 6.2.1).

3.4.2 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductor and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a resin base.

3.4.3 Electromagnetic interference control. When specified (see 6.2.1), the furnaces shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.5 Performance. Each furnace shall be capable of continuous operation at any temperature within its operating range, with loads up to 100 lbs/sq. foot of hearth area.

3.5.1 Heating rate. The Class 1 furnace shall be capable of heating a charge of low carbon steel equivalent to not less than 40 pounds per square foot of hearth area from 70°F to maximum temperature in one hour. Class 2 and Class 3 furnaces shall be capable of heating an equivalent charge to 2200°F in one hour. The charge is based on gross load, and includes the weight of baskets, trays, fixtures, required to retain the workload. The heating rate is determined by loading the charge in the chamber which has been stabilized at the specified temperature.

* 3.5.2 Temperature uniformity. With the furnace charged and the temperature stabilized, the temperature at any point in the effective working zone shall not vary more than ± 25°F from the instrument control point. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within its operating range.

3.6 Components.

3.6.1 Frame. The furnace shall have a structural steel frame, and shall provide a stable support for the work chamber and a maximum load. If legs are used, they shall be braced and provided with bearing plates. The working height of the furnace hearth shall be between 36 and 42 inches above floor level.

3.6.2 Shell. The outer shell of the furnace shall be constructed of steel plate, not less than 3/16 inch in thickness. The shell shall be reinforced with steel members, and securely welded at all joints, except openings required for repair and rebuild shall be provided with suitable gaskets and bolted. The shell shall be of gas-tight construction, and provided with

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suitable threaded connections for operation with a controlled atmosphere of the type specified for each furnace. The front of the furance surrounding the door opening shall be designed and constructed of such material that warpage will be prevented.

3.6.3 Hearth. The hearth shall be designed to permit maximum heat transfer and uniform temperature throughout the work chamber. Furnaces with maximum operating temperatures above 2000°F shall have hearth plates constructed of silicon carbide material. Each furnace hearth shall be capable of supporting a load of not less than 100 pounds of steel per square foot of hearth area without visible deflection. The hearth plates shall be easily removed and replaced with means provided to allow unrestricted movement for expansion and contraction. When specified (see 6.2.1), the furnace hearth shall be provided with slide or roller rails for moving the charge in and out of the work chamber. When required, the chamber shall be provided with open-grid type shelves. Number and arrangement of shelves shall be as specified (see 6.2.1).

3.6.4 Refractory. The lining shall consist of suitable refractory or insulating brick, selected to minimize atmosphere contamination, resist abrasion and spalling and backed up with insulated brick, block or preformed insulation. Overall thickness and quality of the refractory and insulation shall be such that the outside surface temperature of the Class 1 furnace will not exceed 100°F, or the Class 2 furnace 150°F, or the Class 3 furnace 200°F above ambient air after the temperature has stabilized at maximum operating temperature. When mortar is used, it should be of the same basic material as the units being cemented together. Powdered, shredded or granular insulation shall not be used for insulating purposes.

* 3.6.5 Door. Unless otherwise specified, the door shall be of the vertical lift type, and manually operated with suitable counter weights. When required (see 6.2.1), the door shall be pneumatically operated with pressure regulator, cutoff valve and foot control valve furnished. When operating on an air pressure of 80 PSI, the door shall fully open or close in not more than 10 seconds. The door shall be fully insulated with refractory or insulating material, and shall overlap the brick facing of the furnace not less than 2½ inches on all sides. The door shall be provided with jambs and sufficient seals to form a gas-tight unit when closed. The door shall be provided with a non-fogging sight glass for observing the charge without opening the door.

3.6.6 Flame curtain. Unless otherwise specifed (see 6.2.1), a gas-air-flame curtain shall be provided for the door openings, and shall operate automatically by limit switches through solenoid valves to protect the furnace atmosphere when the door is open.

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3.6.7 Heating elements. Heating elements shall be distributed and located to meet the heating requirements and temperature uniformity specified herein. The elements shall be designed and mounted in accordance with the best accepted commercial practice for the particular type of atmosphere and temperature range specified for each furnace. Heating elements shall be fabricated from material that is compatible with the atmosphere and temperature specified for each furnace and shall be capable of continuous operation at any temperature within the operating range of the furnace. Furnaces equipped with silicon carbide or similar elements that require voltage compensation for change in resistance shall be provided with a multiple-tap transformer having not less than 7 steps. An ammeter or a comparable instrument shall be furnished with furnaces having silicon carbide or equivalent type elements to detect changes in resistance. The power density on the heating elements shall not exceed 15 watts per square inch of radiation surface for alloy type elements, and 50 watts for silicon carbide or equivalent elements.

3.6.8 Thermocouples. Furnaces having a temperature rating in excess of 2000°F shall be provided with platinum-rhodium type thermocouples. Furnaces having a temperature rating of 2000°F and lower shall be provided with chromel-alumel type thermocouples. The thermocouples shall be adequately protected from contamination of furnace atmosphere by means of suitable protective tubes. The thermocouples shall conform to ANSI C96.1. Thermocouple lead wires shall be not less than 25 feet in length and of the same material as the thermocouples, or of a material having the same thermoelectrical characteristics at a thermocouple head temperature between 0 degrees and 250°F.

3.6.9 Program control. When required, the furnace shall be provided with a program control system integrated with the instrument controls. Programming requirements such as heating rates, cooling rates, and soak periods shall be as specified (see 6.2.1).

3.6.10 Instruments. Each furnace shall be provided with a single, self-contained temperature indicating-recording-controller type instrument with an accuracy of 0.25 percent or better of full scale reading. Solid state design shall be employed where possible. The instrument shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element.

* 3.6.10.1 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified, the calibrated chart width shall be not less than six inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be one inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart width, and not

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less than seven inches of chart length. When required, the chart width shall not be less than 9 inches and provisions provided for changing chart speed to 4 and 6 inches per hour as specified (see 6.2.1).

3.6.10.2 Temperature controller. Unless otherwise specified, the temperature controller shall be of the time proportioning type control, and shall operate a contactor through an intermediate relay system. When required, the controller shall be of the current proportioning type control, and shall operate either a saturable core reactor or silicon controlled rectifier as specified (see 6.2.1). Either system furnished shall have the necessary adjustments to be capable of maintaining the set control temperature within +2°F. The control system shall provide thermocouple upscale failsafe protection that will shut off the heat source when the chamber temperature exceeds approximately 100°F above maximum.

3.6.10.3 Temperature indicator. The temperature indicator scale shall be approximately the same length as the temperature recorder chart width. The calibrated scale range shall not be less than 100°F above the maximum temperature of the furnace. The increments between successive temperature marks shall be not greater than 20°F, with the spacing of numbered marks not greater than 400°F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

* 3.6.10.4 Excess temperature control. The furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the operating control circuitry. The controller shall provide for selecting the over-temperature control point, and shall function to shut off the heat source to the furnace when the over-temperature control point is reached. In case the furnace is shut down by the excess temperature control system, the system shall sound an audible alarm, and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

3.6.10.5 Instrument panel. Floor mounted instrument panels shall be provided for mounting all instruments and accessory controls. The panels shall conform to JIC Standard EGP-1. All ventilated openings shall be designed to prevent entrance of any deleterious substance. Signal panel lights shall be provided to indicate the furance cycle performance. The panels shall provide flush or semi-flush mounting of all instruments.

3.6.10.6 Thermocouple burn-out alarm signal. The excess temperature control and temperature controller shall be designed and equipped to provide an audible and visible alarm signal in the event of the loss of either or both thermocouple circuits.

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3.7 Optional equipment. When required, optional equipment such as work trays and fixtures, loading and extracting equipment, and flow meters shall be furnished. The equipment required, dimension, and design shall be as specified (see 6.2.1).

3.8 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metal nameplate shall be securely attached to each furnace. The nameplate shall contain the information listed below. If the furnace is a special model, the model designation shall include the model of the basic standard furnace and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's serial number
Manufacturer's model number
Power input characteristics and ratings
Element voltage
Inside dimensions (length, width, height)
Maximum operating temperature (^{o}F)
Compatible atmosphere
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code
U S

3.9 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.10 Workmanship. Workmanship of the furnace and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall comprise the examination in 4.5 and all tests in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the tests in 4.6.1 and 4.6.2 and the inspection in 4.7. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.5 Examination. The furnace and equipment shall be examined for design, dimensions, construction, materials, components, electrical and safety compliance, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. All tests shall be conducted at an ambient temperature of $77^{o}F \pm 18^{o}F$. Test instruments and sensing elements used to perform the test specified for the furnace and instruments should have been checked against a standard potentiometer type instrument and calibrated thermocouple of known accuracy within the previous three months to read accurate within $\pm 2^{o}F$.

4.6.1 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANSI C96.1. The instruments shall be tested through their operating range.

4.6.2 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of JIC Standards.

4.6.3 Electromagnetic interference control testing. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.3 in accordance with MIL-STD-461.

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4.6.4 Operational test. The furnace shall be prepared for operation, and operated in accordance with the manufacturer's handbook of instructions until the refractory and insulation has cured. With the chamber temperature stabilized at 2000°F for the Class 1 furnace and at 2200°F for Class 2 and Class 3 furnaces, the chamber shall be loaded with a charge of iron or steel equivalent to not less than 40 lbs. per square foot of hearth area. At the end of one hour, the chamber temperature shall be within 10°F of the initial temperature. Each safety device shall be tested by simulating a failure that normally would cause the device or system to operate. Each system shall perform its intended function.

4.6.5 Performance test. The furnace shall be charged as specified in 4.6.4, and a temperature survey performed in the effective work zone of the heating chamber. The effective work zone (test zone) is defined as being 1/6 less than the actual cubic volume of chamber air space. A minimum of nine test locations shall be used, with one thermocouple located in each corner of the test zone, and one thermocouple located withing three inches of the control thermocouple. For furnaces with a chamber volume of less than 10 cubic feet, the test may be performed with a minimum of three thermocouples, with one each located at the front, center and rear of the test zone. The survey shall be performed at the minimum and maximum temperature range of Class 1 furnace, and the minimum and at 2200°F of the Class 2 and Class 3 furnace. From the time the heating cycle is initiated until the thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the temperature range being surveyed. After thermal equilibrium is reached, readings shall be taken at 5 minute intervals for 30 minutes. The maximum temperature variation between all points under test shall not vary more than ± 25°F from the instrument control point.

4.6.6 Chamber surface temperature test. The chamber temperature shall be stabilized at maximum operating temperature, and the surface temperature of the chamber and door checked at random locations. The surface temperature shall not exceed the limits specified in 3.6.4, except within 3 inches of through-fastening devices. The Class 2 and Class 3 furnace shall be unloaded before increasing the temperature above the melting point of the charge.

* 4.7 Inspection of packaging. Packaging shall be inspected to determine compliance to the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation, packing and marking. Preservation, packing, and marking shall be in accordance with MIL-F-3296. Level of preservation and packing shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. Furnaces covered by this specification are intended for use in heat treating shops, in such processes as hardening, tempering, normalizing and annealing metal parts, utilizing a controlled manufactured atmosphere.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Class and size required (see 1.2).
c. First article approval, if required (see 3.1).
d. Retort, if required (see 3.2).
e. Type of atmosphere the furnace shall be designed and equipped to use, such as endothermic, exothermic, ammonia gas, etc. (see 3.2).
f. Additional safety and health requirements, if required (see 3.2.1 and 6.5).
g. If furnace is required to be configured in a specific measurement system (US or SI) specify required system (see 3.3.8).
h. Electrical requirements, specify voltage and phase (see 3.4.1).
i. Electromagnetic interference control, if required (see 3.4.3).
j. Hearth roller or slide rails, when required (see 3.6.3).
k. Chamber shelves, specify number and spacing when required (see 3.6.3).
l. Power operated door, when required (see 3.6.5).
m. Flame curtain, if different (see 3.6.6).
n. Program control, and programming requirements when required (see 3.6.9).
o. Recorder chart width, if different (see 3.6.10.1).
p. Controller, specify silicon controlled rectifier or saturable core reactor, if required (see 3.6.10.2).
q. Optional equipment, when required (see 3.7).
r. Nameplate, if different (see 3.8).
s. First article inspection, if different (see 4.3).
t. Quality conformance inspection, if different (see 4.4).
u. Level of preservation, packaging and packing required (see 5.1).

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6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. A first article comprises a preproduction item or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments, regarding arrangement for examination, test and approval of the first article.

6.4 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning gurading the point(s) of operation, and other safety and health requirements should be as specified.

* 6.5 Mercury prohibited. When the use of mercury or its compounds in the furnace and its components and accessories is to be prohibited, procurement documents should include a statement to that effect.

6.6 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content, irrespective of the marginal notation and relation to the previous issue.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH, AS
Air Force - 99

User activities:
Army - AV
Navy - MC, OS
DLA - GS

Preparing activity:
DLA - IP

Project Number:
3424-0096

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

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